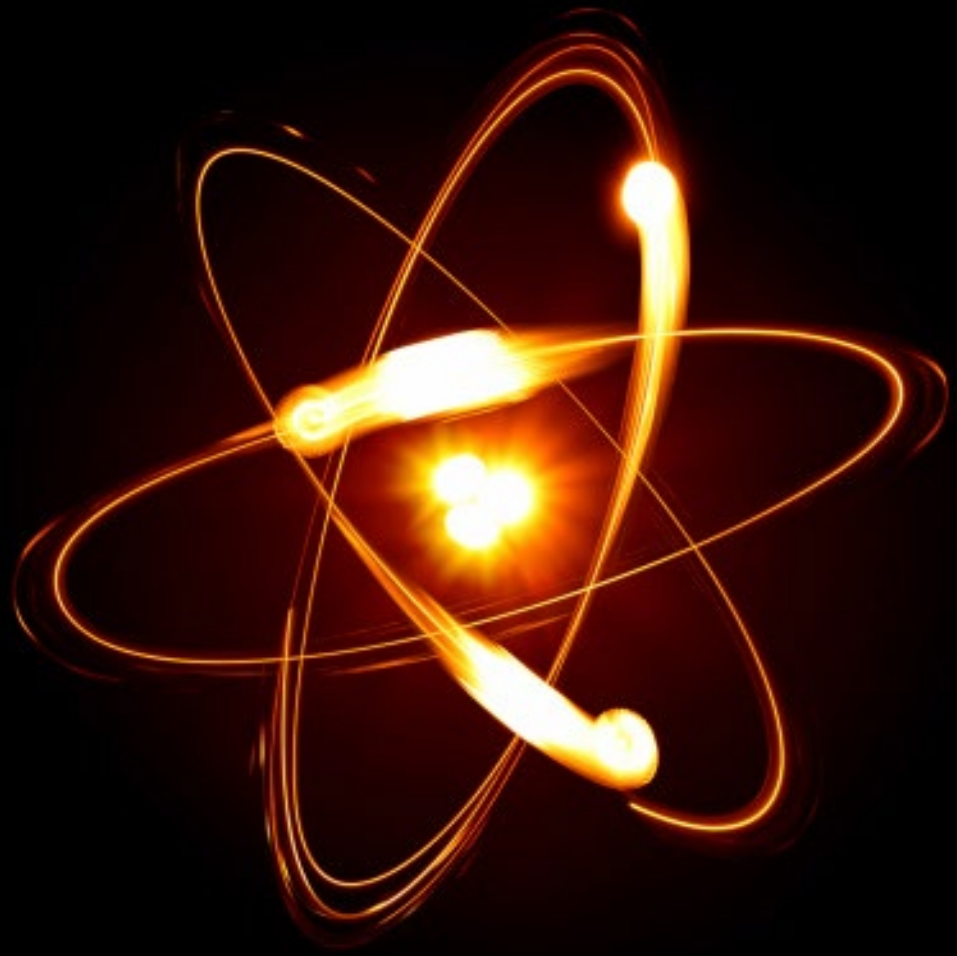
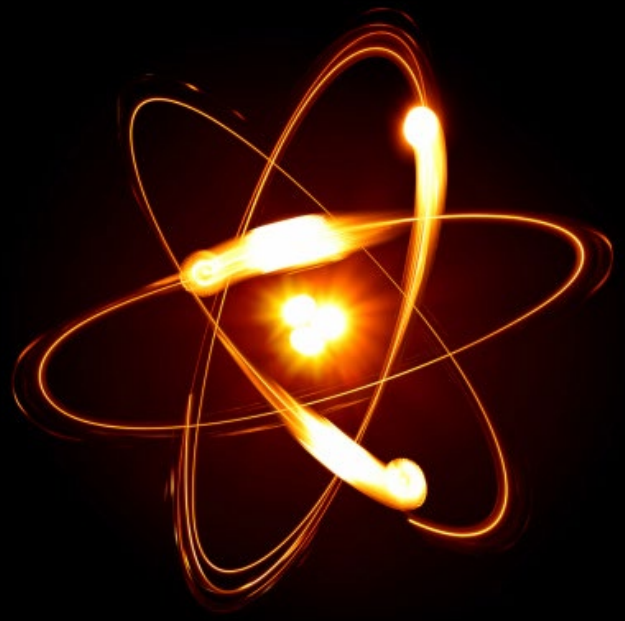
Chapter 21.

Addition of two spins 1/2



Beatriz González del Río

Quantum Physics



Chapter 21.

Addition of two spins 1/2

- Introduction
- Tensor product
- Total angular momentum in classical mechanics
- Total angular momentum in quantum mechanics
- Addition of two spins ½. Elementary method
  - Statement of the problem
  - Eigenvalues of S_z
  - Diagonalization of S^2
  - Results: triplet and singlet

Introduction

Consider an electron in a hydrogen atom, in a state with $l \neq 0$. It has two angular momenta, one associated with the orbital movement (because $l \neq 0$) and another one associated to spin.

How can one construct the total angular momentum of the electron?

"Simply" by adding the two of them.

The quotes are due to the fact that adding two angular momenta is not so simple.

- The first problem arises because usually the two operators act in different spaces.
  For the case just considered $\vec{L}$ acts in configuration space (on orbital variables) while $\vec{S}$ acts in spin space (on spin variables).

  For a two electron atom one angular momentum corresponds to one electron (acts on that electron's variables) while the second corresponds to the other electron (acting on the other electron's variables).

  How do we add operators that act in different state spaces?

Introduction

- The second problem arises because we know that the eigenvalues of an angular momentum must satisfy certain relations. How can we then compute the value(-s) of j associated to the sum of two angular momenta with j-values j_1 and j_2 ?

We first tackle the first question through the tensor product of spaces.

Then we address the possible eigenvalues of the angular momentum given by a sum of two angular momenta.

Tensor product[3]

Tensor product space

Consider two systems whose state spaces are $\mathcal{E}_1$ and $\mathcal{E}_2$. If we consider both systems together as forming one global system, how is the state space of this global system constructed?

It is given by the tensor product of the two spaces:

$$\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$$

Properties of the tensor product of two vectors:

- Linear with respect to multiplication by a scalar:

$$[\lambda|\phi(1)\rangle] \otimes |\psi(2)\rangle = \lambda\big[|\phi(1)\rangle \otimes |\psi(2)\rangle\big]$$

$$|\phi(1)\rangle \otimes [\mu|\psi(2)\rangle] = \mu\big[|\phi(1)\rangle \otimes |\psi(2)\rangle\big]$$

[3]Cap II section F Cohen

Tensor product

- Distributive with respect to vector sum:

$$|\phi(1)\rangle\otimes[|\psi_1(2)\rangle + |\psi_2(2)\rangle] = |\phi(1)\rangle\otimes|\psi_1(2)\rangle + \left|\phi(1)\rangle\otimes|\psi_2(2)\right\rangle$$

- Given a basis set in $\mathcal{E}_1$ $\{|u_i(1)\rangle\}$ and a basis set in $\mathcal{E}_2$ $\left\{\left|v_j(2)\right\rangle\right\}$, $\left\{|u_i(1)\rangle\otimes\left|v_j(2)\right\rangle\right\}$ is a basis set in $\mathcal{E}$. $\forall|\Psi\rangle \in \mathcal{E}$

$$|\Psi\rangle = \sum_{i,j} c_{i,j}|u_i(1)\rangle\otimes\left|v_j(2)\right\rangle = \sum_{i,j} c_{i,j}\left|u_i(1), v_j(2)\right\rangle$$

- In $\mathcal{E}$ there are vectors that are not a tensor product of a vector in $\mathcal{E}_1$ and a vector in $\mathcal{E}_2$, but these vectors (in fact all of the vectors of $\mathcal{E}$) can be expressed as linear combinations of such type of vectors (those of the basis set).

Inner product in $\mathcal{E}$

Consider two states of $\mathcal{E}$ that are tensor products of vectors of $\mathcal{E}_1$ and $\mathcal{E}_2$, $|\phi(1)\rangle\otimes|\psi(2)\rangle$ and $|\phi'(1)\rangle\otimes|\psi'(2)\rangle$

We define their inner product as:

$$(\langle\phi(1)|\otimes\langle\psi(2)|)\,(|\phi'(1)\rangle\otimes|\psi'(2)\rangle) = \langle\phi(1)|\phi'(1)\rangle\,\langle\psi(2)|\psi'(2)\rangle$$

What about states that are not tensors products? We write them in terms of those of the basis (which are tensors product), and apply linearity and the previous definition.

$$|\psi\rangle, |\phi\rangle \in \mathcal{E} \qquad |\psi\rangle = \sum_{ij} c_{ij}|u_i(1)\rangle\otimes|v_j(2)\rangle \qquad |\phi\rangle = \sum_{ij} d_{ij}|u_i(1)\rangle\otimes|v_j(2)\rangle$$

$$\langle\psi|\phi\rangle = \sum_{ij}\sum_{kl} c_{ij}^* d_{kl}\big(\langle u_i(1)|\otimes\langle v_j(2)|\big)\,(|u_k(1)\rangle\otimes|v_l(2)\rangle)$$

Operators in $\mathcal{E}$

Let $A(1)$ be an operator acting in $\mathcal{E}_1$

How can we define the action of $A(1)$ on a vector of $\mathcal{E}$?

$$A(1)[|\phi(1)\rangle \otimes |\psi(2)\rangle] = [A(1)|\phi(1)\rangle] \otimes |\psi(2)\rangle$$

For an operator $B(2)$ acting in $\mathcal{E}_2$ we follow a similar procedure:

$$B(2)[|\phi(1)\rangle \otimes |\psi(2)\rangle] = |\phi(1)\rangle \otimes [B(2)|\psi(2)\rangle]$$

How do these operators act on vectors that are not tensor products? Write the vector in terms of those of the basis and apply linearity and the previous definition. $|\psi\rangle \in \mathcal{E}$ with $|\psi\rangle = \sum_{ij} c_{ij} |u_i(1)\rangle \otimes \left|v_j(2)\right\rangle$

Then

$$A(1)|\psi\rangle = \sum_{ij} c_{ij} A(1)|u_i(1)\rangle \otimes \left|v_j(2)\right\rangle$$

Total angular momenta in classical mechanics

System of N classical particles

Total angular momentum of the system with respect to a fixed point O is

$$\vec{\mathcal{L}} = \sum_{i=1}^{N} \vec{\mathcal{L}}_i \qquad\qquad \vec{\mathcal{L}}_i = \vec{r}_i \times \vec{p}_i$$

Now, the time derivative is $\frac{d\vec{\mathcal{L}}}{dt} = \sum_{i=1}^{N} \vec{r}_i \times \vec{F}_i^{ext}$ which is the momentum, with respect to point O, of the **external** forces.

If $\vec{F}_i^{ext}$ are zero (isolated system) OR all directed towards O, then $\frac{d\vec{\mathcal{L}}}{dt} = 0$ which implies that $\vec{\mathcal{L}}$ is a constant of the motion.

But if there are internal forces, then

$$\frac{d\vec{\mathcal{L}}_i}{dt} \neq 0 \qquad \text{even though} \qquad \frac{d\vec{\mathcal{L}}}{dt} = 0$$

Conclusion: In a system of interacting particles, only the **total angular momentum** is a constant of the motion. It highlights the importance of the $\vec{\mathcal{L}}$.

Total angular momenta in quantum mechanics

Consider two non-interacting particles:

Total Hamiltonian ($V(r_i)$ are **central** potentials) $H_0 = H_1 + H_2$ $H_i = -\frac{\hbar^2}{m_i}\nabla_i^2 + V(r_i)$ $(i = 1,2)$

We know that ($\vec{L}_i$ is the angular momentum of particle i)

$$[\vec{L}_1, H_1] = [\vec{L}_1, H_2] = 0 \qquad [\vec{L}_2, H_2] = [\vec{L}_2, H_1] = 0$$

and **the $\vec{L}_i$ (i.e. L_{ix}, L_{iy}, L_{iz}) are constants of the motion.**

Assume that the two particles interact by a potential $v(|\vec{r}_1 - \vec{r}_2|)$. Now the total hamiltonian becomes

$$H_0 = H_1 + H_2 + v(|\vec{r}_1 - \vec{r}_2|)$$

and the commutation relations change $[\vec{L}_1, H] = [\vec{L}_1, v(|\vec{r}_1 - \vec{r}_2|)] \neq 0$ $[\vec{L}_2, H] = [\vec{L}_2, v(|\vec{r}_1 - \vec{r}_2|)] \neq 0$

and **the $\vec{L}_i$ are NO LONGER constants of the motion.**

But, if we consider $\vec{L} = \vec{L}_1 + \vec{L}_2$ then

$$[\vec{L}, H] = 0 \qquad \text{that is:} \qquad [L_x, H] = 0 \qquad [L_y, H] = 0 \qquad [L_z, H] = 0$$

and $\vec{L}$ **is a constant of the motion.**

Total angular momenta in quantum mechanics

Single particle with spin

Consider a particle subjected to a central potential, $V(r)$, only (i.e. the electron in a hydrogen atom)

The three components of the orbital angular momentum (L_x, L_y and L_z) commute with its hamiltonian

If we now take into account the spin, we know that the spin operator commutes with the orbital operators, including the hamiltonian, which means that S_x, S_y and S_z are constants of the motion.

But... if we also consider the relativistic corrections... then there appears an additional term in the hamiltonian, namely the *spin-orbit* coupling term

$$H_{SO} = \xi(r)\vec{L}\cdot\vec{S}$$

Now, neither $\vec{L}$ nor $\vec{S}$ commute with the **total** hamiltonian

$$[L_z, H_{so}] = \xi(r)\left[L_z, L_xS_x + L_yS_y + L_zS_z\right] = \xi(r)\left(i\hbar L_yS_x - i\hbar L_xS_y\right)$$

$$[S_z, H_{so}] = \xi(r)\left[S_z, L_xS_x + L_yS_y + L_zS_z\right] = \xi(r)\left(i\hbar L_xS_y - i\hbar L_yS_x\right)$$

Total angular momenta in quantum mechanics

But, if we consider $\vec{J} = \vec{L} + \vec{S}$ (**total angular momentum**) then, (J_x, J_y, J_z) commute with the total hamiltonian.

$$[J_z, H_{so}] = [L_z + S_z, H_{so}] = 0$$

and the same happens for the other two components.

Summary:

- We had two partial angular momenta, $\vec{J}_1$ and $\vec{J}_2$, which commute.
- We know a basis of the (total) state space made of eigenvectors common to $J_1^2, J_{1z}, J_2^2, J_{2z}$
- But $\vec{J}_1$ and $\vec{J}_2$ are NOT constants of the motion.
- However, if we consider $\vec{J} = \vec{J}_1 + \vec{J}_2$, this new observable commutes with the hamiltonian.

Addition of angular momenta

Objective:

- To construct a new basis, made of eigenvectors of J^2 and J_z (Problem of addition of two angular momenta)
- To find the eigenstates of H, it is much simpler (and convenient) to work in the basis made of eigenvectors of J^2 and J_z.
- Now, as H commutes with both J^2 and J_z, then its matrix can be decomposed into as many blocks as there are eigensubspaces associated with the different sets of eigenvalues of J^2 and J_z.

As an application of the previous theory, let us consider the simple case in which the two partial angular momenta we want to add are two spin 1/2.

Addition of two spins 1/2[4]

We consider two spin 1/2 particles, with spin operators $\vec{S}_1$ and $\vec{S}_2$ respectively.

Let $\mathcal{E}_1$ be the spin state space of electron 1. A basis of this space is

$$\{|s = 1/2\,, m_s = 1/2\rangle, |s = 1/2\,, m_s = -1/2\rangle\} \equiv \{|+\rangle, |-\rangle\}$$

Let $\mathcal{E}_2$ be the spin state space of electron 2, where we also consider the basis $\{|+\rangle, |-\rangle\}$.

Let $\mathcal{E}$ be the spin state space of the two electrons $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$. A basis of this space is:

$$\{|+\rangle\otimes|+\rangle, |+\rangle\otimes|-\rangle, |-\rangle\otimes|+\rangle, |-\rangle\otimes|-\rangle\} \equiv \{|+,+\rangle, |+,-\rangle, |-,+\rangle, |-,-\rangle\} \equiv \{|\varepsilon_1, \varepsilon_2\rangle\}$$

where $\varepsilon = \pm$.

[4]Chap X section B Cohen

Addition of two spins 1/2

By construction, the basis vectors are eigenstates of $S_1^2, S_2^2, S_{1z}, S_{2z}$

- $S_1^2|\varepsilon_1,\varepsilon_2\rangle = S_1^2(|\varepsilon_1\rangle\otimes|\varepsilon_2\rangle) = (S_1^2|\varepsilon_1\rangle)\otimes|\varepsilon_2\rangle = \left(\frac{3}{4}\hbar^2|\varepsilon_1\rangle\right)\otimes|\varepsilon_2\rangle = \frac{3}{4}\hbar^2|\varepsilon_1,\varepsilon_2\rangle$

- $S_2^2|\varepsilon_1,\varepsilon_2\rangle = \frac{3}{4}\hbar^2|\varepsilon_1,\varepsilon_2\rangle$
- $S_{1z}|\varepsilon_1,\varepsilon_2\rangle = \varepsilon_1\frac{\hbar}{2}|\varepsilon_1,\varepsilon_2\rangle$
- $S_{2z}|\varepsilon_1,\varepsilon_2\rangle = \varepsilon_2\frac{\hbar}{2}|\varepsilon_1,\varepsilon_2\rangle$

S_1^2, S_2^2, S_{1z} and S_{2z} constitute a CSCO

Total spin. The total spin of the two-electron system is defined as:

$$\vec{S} = \vec{S}_1 + \vec{S}_2$$

so,

$$S_x = S_{1x} + S_{2x} \qquad S_y = S_{1y} + S_{2y} \qquad S_z = S_{1z} + S_{2z}$$

Addition of two spins 1/2

Let us show that $\vec{S}$ is an angular momentum operator. Evaluate the commutation relations.

$$[S_x, S_y] = [S_{1x} + S_{2x}, S_{1y} + S_{2y}] \overset{(*)}{=} i\hbar S_z$$

$$[S_y, S_z] = [S_{1y} + S_{2y}, S_{1z} + S_{2z}] = i\hbar S_x$$

$$[S_z, S_x] = [S_{1z} + S_{2z}, S_{1x} + S_{2x}] = i\hbar S_y$$

$$[S^2, S_x] = [S^2, S_y] = [S^2, S_z] = 0$$

Note that S^2 DOES NOT commute with either S_{1z} or S_{2z}.

$$(*)\ [S_{1x} + S_{2x}, S_{1y} + S_{2y}] = S_{1x}S_{1y} + S_{1x}S_{2y} + S_{2x}S_{1y} + S_{2x}S_{2y} - S_{1y}S_{1x} - S_{1y}S_{2x} - S_{2y}S_{1x} - S_{2y}S_{2x} =$$

$$= [S_{1x}, S_{1y}] + [S_{2x}, S_{2y}] = i\hbar S_z$$

Addition of two spins 1/2

We obviously have $S_z = S_{1z} + S_{2z}$. It will also be useful to write S^2 in terms of the 1-particle operators:

$$S^2 = (\vec{S}_1 + \vec{S}_2)(\vec{S}_1 + \vec{S}_2) = S_1^2 + S_2^2 + \vec{S}_1\vec{S}_2 + \vec{S}_2\vec{S}_1 = S_1^2 + S_2^2 + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+}$$

The basis of $\mathcal{E}$, $|\varepsilon_1\varepsilon_2\rangle$, is made up of common eigenstates of $S_1^2, S_2^2, S_{1z}, S_{2z}$.

But in terms of the previous expressions it is very easy to see that the operators S_1^2, S_2^2, S^2 and S_z commute with each other, and consequently it is possible to find a basis of $\mathcal{E}$ with eigenstates common to them. Let these states be $\{|\mathcal{S}, \mathcal{M}\rangle\}$.

Since $\vec{S}$ is an angular momentum we will have

$$S^2|\mathcal{S}, \mathcal{M}\rangle = \mathcal{S}(\mathcal{S}+1)\hbar^2|\mathcal{S}, \mathcal{M}\rangle \qquad S_z|\mathcal{S}, \mathcal{M}\rangle = \mathcal{M}\hbar|\mathcal{S}, \mathcal{M}\rangle$$

Besides $S_1^2|\mathcal{S}, \mathcal{M}\rangle = S_2^2|\mathcal{S}, \mathcal{M}\rangle = {}^3/_4\,\hbar^2|\mathcal{S}, \mathcal{M}\rangle$

In order to find them we compute the matrices that represent S_z and S^2 in the old basis and diagonalize them together.

Addition of two spins 1/2

From $\{S_1^2, S_2^2, S_{1z}, S_{2z}\} \Longrightarrow \{S_1^2, S_2^2, S^2, S_z\}$

From eigenvectors $\{|1/2, 1/2, \varepsilon_1, \varepsilon_2\rangle\} \Longrightarrow \{|1/2, 1/2, \mathcal{S}, \mathcal{M}\rangle\}$

Matrix that represents S_z in the (old) basis $|\varepsilon_1, \varepsilon_2\rangle$:

$$S_z|\varepsilon_1, \varepsilon_2\rangle = (S_{1z} + S_{2z})|\varepsilon_1, \varepsilon_2\rangle = \left(\frac{\hbar}{2}\varepsilon_1 + \frac{\hbar}{2}\varepsilon_2\right)|\varepsilon_1, \varepsilon_2\rangle$$

therefore,

$$S_z = \hbar \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

This matrix is diagonal, and therefore the possible values of $\mathcal{M}$ are 1, 0, 0 -1 (The eigenvalue $\mathcal{M} = 0$ is 2-fold degenerate).

Addition of two spins 1/2

Matrix that represents S^2 in the basis $|\varepsilon_1, \varepsilon_2\rangle$:

$$S^2|+,+\rangle = (S_1^2 + S_2^2 + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+})|+,+\rangle = \left(\frac{3\hbar^2}{4} + \frac{3\hbar^2}{4} + 2\frac{\hbar}{2}\frac{\hbar}{2} + 0 + 0\right)|+,+\rangle = 2\hbar^2|+,+\rangle = \hbar^2\begin{pmatrix}2\\0\\0\\0\end{pmatrix}$$

$$S^2|+,-\rangle = (S_1^2 + S_2^2 + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+})|+,-\rangle = \left(\frac{3\hbar^2}{4} + \frac{3\hbar^2}{4} + 2\frac{\hbar}{2}\left(\frac{-\hbar}{2}\right)\right)|+,-\rangle + 0 + \hbar\hbar|-,+\rangle =$$

$$= \hbar^2|+,-\rangle + \hbar^2|-,+\rangle = \hbar^2\begin{pmatrix}0\\1\\1\\0\end{pmatrix}$$

Addition of two spins 1/2

$$S^2|-,+\rangle = (S_1^2 + S_2^2 + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+})|-,+\rangle = \hbar^2|-,+\rangle + \hbar^2|+,-\rangle = \hbar^2 \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$S^2|-,-\rangle = (S_1^2 + S_2^2 + 2S_{1z}S_{2z} + S_{1+}S_{2-} + S_{1-}S_{2+})|-,-\rangle = \left(\frac{3\hbar^2}{4} + \frac{3\hbar^2}{4} + 2\frac{\hbar}{2}\frac{\hbar}{2} + 0 + 0\right)|-,-\rangle = 2\hbar^2|-,-\rangle = \hbar^2 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 2 \end{pmatrix}$$

Addition of two spins 1/2

Finally

$$S^2 = \hbar^2 \left(\begin{array}{c|cc|c} 2 & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 2 \end{array}\right)$$

The matrix is block-diagonal

- $|+,+\rangle$ and $|-,-\rangle$ are eigenstates of S^2 with a value $S = 1$ (eigenvalue $2\hbar^2$)
- We then diagonalize the matrix $\hbar^2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ (in the subspace generated by $|+,-\rangle, |-,+\rangle$).
- The eigenvalues of this matrix are $2\hbar^2$ and 0, so $S = 1,0$
- The corresponding eigenstates are

$$\frac{1}{\sqrt{2}}[|+,-\rangle + |-,+\rangle] \qquad \text{and} \qquad \frac{1}{\sqrt{2}}[|+,-\rangle - |-,+\rangle]$$

Addition of two spins 1/2

Summing up:

Eigenvalues: $\mathcal{S} = 1, 0$

Eigenstates:

$$|0,0\rangle = \frac{1}{\sqrt{2}}[|+,-\rangle - |-,+\rangle] \qquad \text{Singlet}$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}[|+,-\rangle - |-,+\rangle] \qquad |1,1\rangle = |+,+\rangle \qquad |1,-1\rangle = |-,-\rangle \qquad \text{Triplet}$$

- The four vectors $|\mathcal{S},\mathcal{M}\rangle$ form an orthonormal basis. The set $\{S_1^2, S_2^2, S^2, S_z\}$ constitute a CSCO.
- The three states of the triplet are symmetric under an exchange of the particles, while singlet state is antisymmetric.
- When two spin 1/2's ($s_1 = s_2 = 1/2$) are added, the number $\mathcal{S}$ which characterizes the eigenvalues $\mathcal{S}(\mathcal{S}+1)\hbar^2$ of the observable S^2 takes the values 1, 0.
- With each of the two $\mathcal{S}$'s values, is associated a family of $\mathcal{S}(\mathcal{S}+1)$ values of $\mathcal{M}$